



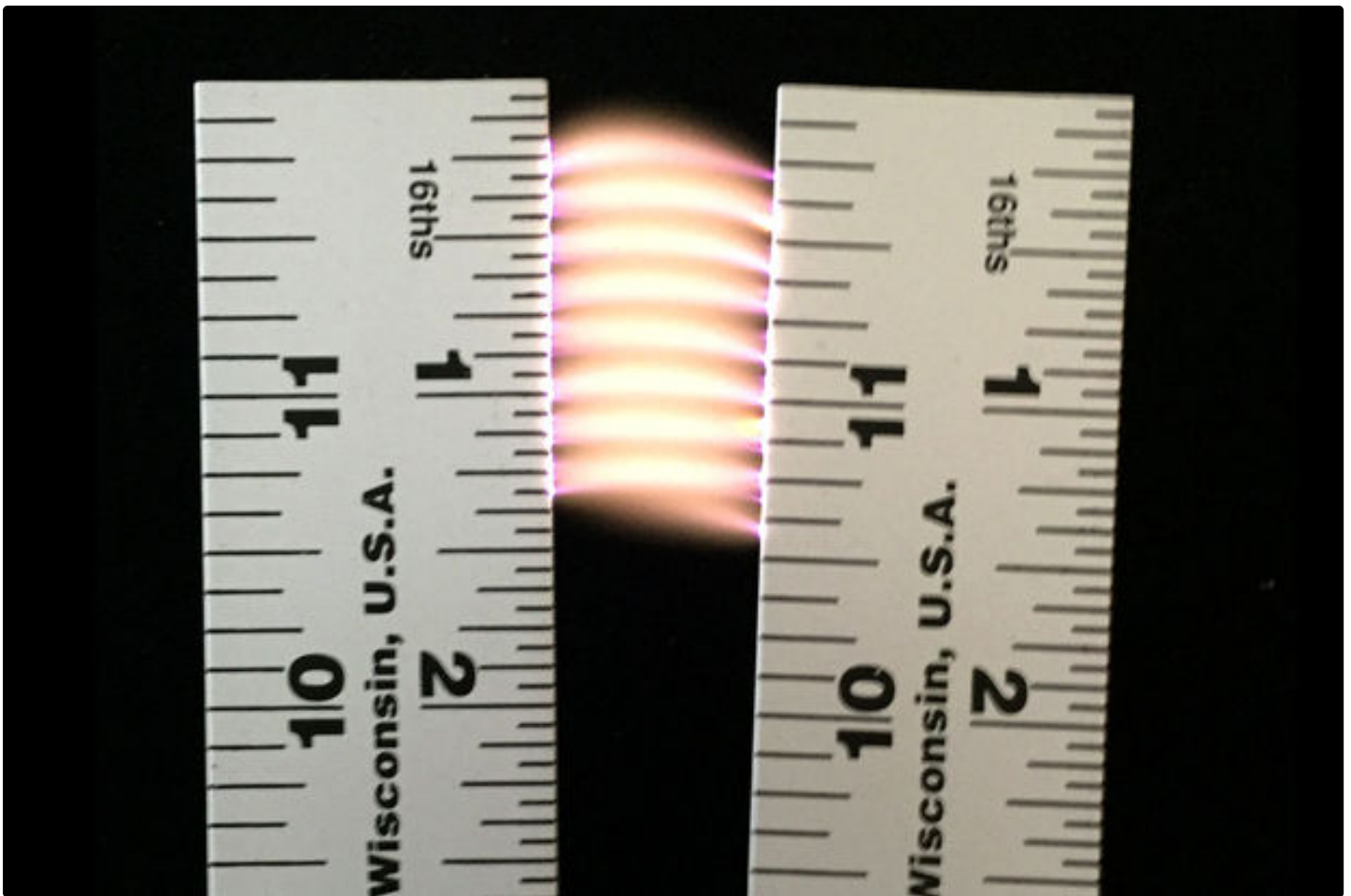
instructables classes

ELECTRONICS CLASS (/class/Electronics-Class/)

LESSON 3: ELECTRICITY

with randofu

There is nothing remarkably shocking about electricity except electricity itself. If you are reading this on a computer, you have likely been making use of electricity your whole life. Most of the objects and systems you encounter are in some way made possible by electricity; whether through the machines that produce them, the machines that facilitate the organization of goods and people, or the the object itself. Machines that run on electricity are so ubiquitous, it is important to understand it. Bear with me as we go over some of the more important concepts.



Charge and Ground



Electricity consists of a charge, which is the build up of positively or negatively charged electrons.

Ground is an area that neutralizes that charge by removing excess charge.

For instance, charged electrons build up in a thundercloud. When it reaches a certain threshold of charge, the electricity is able to ionize the air and arc across the sky as a plasma. Albeit it could travel in any direction, it goes towards the Earth because of its large difference in charge between the lightning bolt and the ground. Just like the north and south poles of a magnet attract, so to do negative and positive charges.

However, unlike magnets which, keep their charge when they meet, electrical charge functions differently. When the (typically) negative charge from the cloud strikes the much larger positively charged surface of the Earth, the Earth is able to absorb and dissipate the negatively charged particles from the thunder cloud. This is what is meant by electric ground neutralizing the charge. The same thing that happens during a storm also happens on a much smaller scale with all electronics.

Static Vs. Current

There are two types of electricity, static and current.



Static electricity is a buildup of electrons that creates an imbalance and sudden discharge. This is the type of electrical charge that lightning is, and that you get when you rub your hair against a balloon.

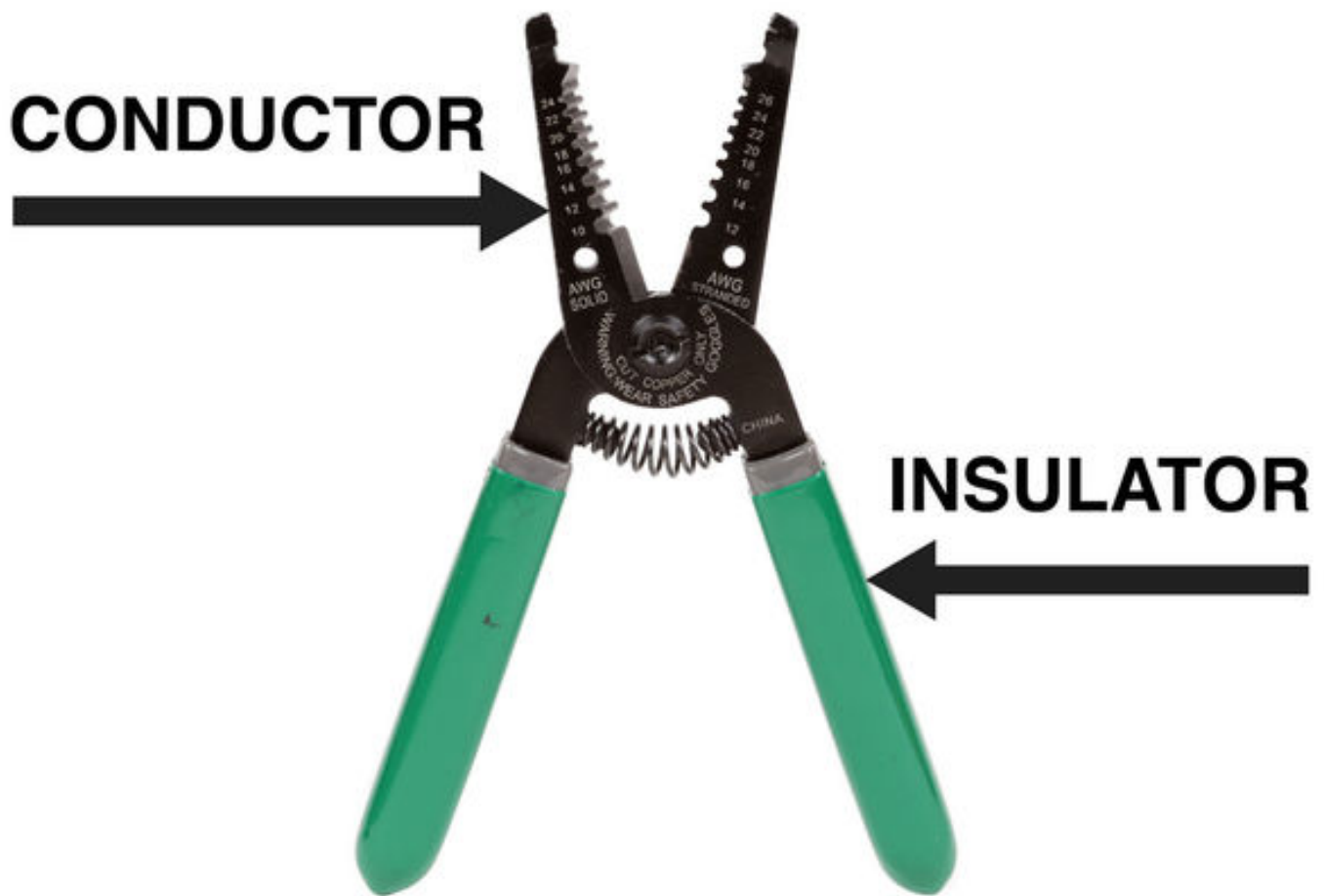
Current electricity is a constant flow of electrons between a source and ground. This is the type of electricity that power electrical devices. Most importantly, it enables you to make waffles.

In this class we will entirely be dealing with electrical current.

Conductors Vs. Insulators

To put it simply, an electrical conductor allows the flow of electricity through it. Typically electrical conductors are metal.

An insulator prevents the flow of electricity through it. Typically insulators are materials like rubber, wood, and plastic.



AC / DC

When we are talking about AC/DC, we are not talking about the Australian hard rock band, but the two different types of electrical current.

◦

There is alternating current or AC. It is called alternating current because the electrical signal alternates above and below the electrical ground (0V). So, if you were to look at the picture of the 12V AC waveform above, you will notice that it alternates above and below the electrical ground (indicated by the little yellow marker on the left). This type of waveform consists of a current that is constantly fluctuating between a positive and a negative voltage.

◦

AC electricity is the type of current you will find coming out of your home wall socket. This is because AC electricity is easier to transmit over long distances than other types.

◦

The other type of current is direct current or DC. It is so named because it travels in a direct line above ground. If you look at the waveform of a 12V DC voltage supply, you will notice that it's basically a solid yellow line running parallel above the ground. This type of electricity consists of a steady positive voltage, set apart from a ground plane.



DC electricity is the type of current that comes from batteries, which are basically special containers that store a predetermined amount of voltage.

◦

AC wall current can also be rectified (more on this later) to become DC voltage. Just because a device plugs into the wall does not necessarily make it an AC device. By reading the label on the plug, you can determine what the output power is.

Aside from just voltage and current, if you look carefully at the bottom of the label, you will also see an illustration that even tells you the inside of the plug is a positive DC voltage and the outer barrel is ground.

Voltage and Current

Imagine a ball being thrown through the air.

◦

Voltage is the speed the ball is traveling. This is measured in Volts. It's symbol is a capital V.

◦

Current is the size of the ball being thrown. This is measured in Amperes (or Amps for short). Its symbol is a capital A.

◦

All DC electricity can be thought of as having a voltage and a current. In order to determine what these are, you can use a multimeter.

◦

To measure voltage, turn the dial to the V with with the straight and dashed lines (not the squiggle). This is symbol for DC voltage, which we will primarily be working with throughout this course. Plug the red probe into the volt socket. This is labeled V and/or is red. Plug the black wire into the ground socket. This will either be labeled ground, COM, and/or is black.

◦

It might also have the ground symbol drawn next to it which looks like a dumbbell that has broken in half and fallen on its side.

◦

Place the red probe on the positive terminal of the battery. Place the black probe on ground (or "minus") terminal of the battery. You should get a reading around 1.5V.

◦

If you reverse the probes of the multimeter you will notice that the meter will give you a negative voltage reading. The reason for this is that DC electricity has a positive voltage and a ground voltage.

You can determine a voltage by subtracting the voltage at the red probe (presumably positive) from the voltage at the black probe (presumably ground). So, if when power and voltage are read correctly the formula is:

◦

However, when you touch the red wire to ground and the black wire to the positive voltage, your formula actually becomes:

◦

The reason for this is that DC electricity has a polarity where one side is always a positive voltage, and one side (ground) is always 0. When you measure the electricity backwards, you get a backwards reading. Thus, if you get a negative voltage reading, you are measuring backwards! Reverse your probes.

◦

To measure current, unplug the red probe from the voltage socket, and connect it into the socket rated for Amps (notice how it says "10A MAX" in the photo above). Next, turn the dial to the A followed by the straight and dashed line. This is the current symbol. If you need to pick a range for the current you

are reading, the middle setting is a good place to start.

Measuring current is a bit more tricky because to measure current something must be drawing some (i.e. using the power source). The amount of current flowing from a battery differs depending on what is connected to it. Some things like a piezo buzzer (pictured left) might draw only a little bit of current. Other things like a DC motor (picture right) might draw a lot of current from the battery.

For instance, to see how much current a motor is drawing, we disconnect the motor's power wire from the battery and connect the multimeter in-between. In other words, the multimeter's red probe gets connected to the positive terminal on the battery, and the black probe gets connected to the red wire on the motor. Now when the motor is powered up, the current will pass through the multimeter and we will be able to read it. In this example, the motor is drawing 70 mA (milliamps).



Ohm's Law

Before I dive into ohm's law, let's revisit our ball analogy. If you have a small ball traveling at a very high speed, it could potentially have as much or more power as a large ball traveling at a very low speed. In this way, you could say there is a direct relationship between the speed a ball is traveling, the

size of the ball, and the potential power of the ball.

◦

Of course though, we are actually not really talking about balls, but electricity. When dealing with electricity, voltage and current are in a direct relationship with power. In a circuit, power is expressed in terms of Watts. The symbol for this is a W.

◦

You likely have heard the term Watts before in relation to light bulbs. All light bulbs are measured in the amount of power they consume.

◦

If we know how many watts a light bulb can consume, and that it is receiving 120 volts from a wall outlet, we can then deduce how much current it will require. By dividing 60W by 120V, we are left with the value of 0.5 amperes of power. On account of this relationship between the three, given any two values, we can solve for the third.

◦

There is also another factor we have yet to talk about that also plays a role, and that is resistance. In our analogy, resistance is the headwind that the ball must fight against to move forwards. On a calm day, there might be little resistance to its flight, but on a windy day, it might have to fight against the wind pretty hard. Again, we are actually talking about electrical resistance in a circuit and not throwing a ball.

◦

Resistance pushes against the flow of electricity. As such, it is also in direct relationship with Watts, Voltage and Current. Resistance is expressed in Ohms (after it's discoverer). This mathematical relationship between Watts, Voltage, Current and Resistance is unsurprisingly called Ohm's Law.

◦

Ohm's Law is not something you must memorize, but it will play an important role later when determining how much resistance a circuit ***must*** have. Thanks to this law, a circuit having a minimum amount of resistance is not optional, but necessary. The energy in the circuit must encounter resistance in order to expend itself. The thing in the circuit which uses energy is considered the Load. If an electrical supply is connected to ground without a load to use up the energy, bad things will happen; but more on that in a bit.

◦

Also, when you encounter staggered lines in a schematic, they represent a battery power supply. Theoretically each pair of long and short lines represents one battery cell. For example, this rendering

stands for three batteries in series. However, people play fast and loose when drawing batteries into schematics and this is not always the case. Look for the listed supply voltage.



Series and Parallel

When building a circuit, there are basically two ways in which you can connect a load. They can either be in series or parallel.

◦

When you wire electrical components in series, they are in a line.

So, if you were to break the circuit in any way -- say by unplugging a light bulb -- electricity has no path to continue flowing and the whole connection is broken.

◦

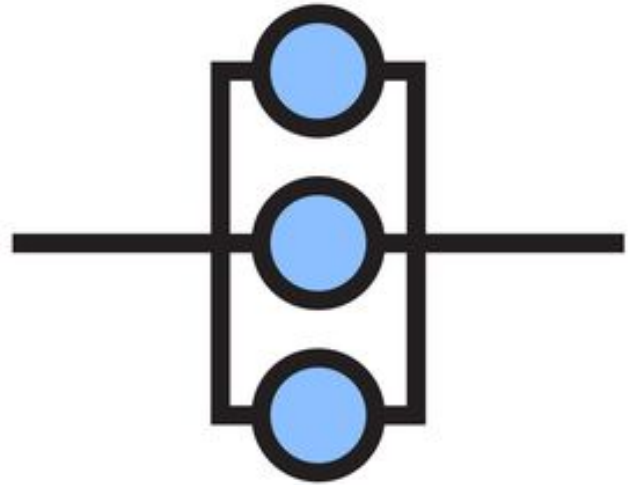
Alternately, you can prevent this problem by wiring the light bulbs in parallel. When wired in this way, they are connected side-by-side and share the same electrical connections on both ends.

Thus, when you unplug one bulb, electricity can continue flowing through the other one unfettered.

SERIES



PARALLEL



All About Batteries

Power supplies such as batteries can also be connected in series and parallel. However, before we get to that, let us take a moment and talk a bit more about batteries.

The most common types of batteries you will encounter are standard cylindrical dry cell batteries. Most notably these consist of AAA, AA, C, and D batteries. What is important to know about these batteries, is that even though they are different sizes, they are all rated at 1.5V (remember - V is the abbreviation for volts).

What changes as they get bigger in size is the amount of power they are capable of producing. A D battery will be able to provide power for much longer than a AAA battery. In other words, a bigger battery can provide more amperes for a longer amount of time than a smaller battery.

Batteries are measured in Amp-Hours or Ah. This is basically the measure of how many amperes can be drawn from the battery in an hour. For instance, a 20Ah battery will let you draw 1 ampere for 20 hours. However, let's say you are building a giant robot and it needs 5 amperes per hour; in this case you can run that robot for about 4 hours using the same battery until it runs out ($20\text{Ah} / 5\text{A} = 4\text{ hours}$).

It may have by now dawned upon you that 1.5V is not very much. You might be wondering why we don't just use a 9V battery instead? Assuredly a 9V battery is producing more power than a 1.5V battery.

This, in fact, is not true. 9V batteries actually are not great in producing power at all. A good way to think of a 9V battery is to imagine 6 really small 1.5V batteries smushed together inside. In fact, if you take a 9V apart, that is essentially what you will find inside. Now, compare that to the size of 6 AA batteries for instance. The 9V batteries are rather tiny by comparison!

The only time 9V batteries come in handy is when you need a relatively high voltage for a project that doesn't require a lot of current and it needs to fit it into a small enclosure. For example, it is a great battery for the guitar pedal we are going to be building in the transistor project.

◦

Well then, you may be wondering how you can power anything if batteries are only 1.5V? The answer is rather simple. We connect them in series. What this means is that we connect them front-to back in a row. So the positive (plus) end of one battery gets connected to the ground (minus) end of the next battery, and so on, and so forth. We can then calculate the new voltage simply by adding 1.5V for each battery in the series. So, if you have three 1.5V batteries in series you could multiply 1.5V times three to get a total of 4.5V.

◦

The easiest way to connect batteries in series is to use a battery holder. To calculate the voltage that a battery holder provides, you simply need to count the number of batteries it holds and multiply it by 1.5V, just as you would with any other set of batteries connected in series. So, a 4-cell battery holder would produce 6V of power ($1.5V \times 4 = 6V$).

◦

To access the power provided, simply connect the red wire to the positive terminal on your project, and the black wire to ground. As a reminder from earlier, red wires always indicate a positive voltage, and black always indicate ground.

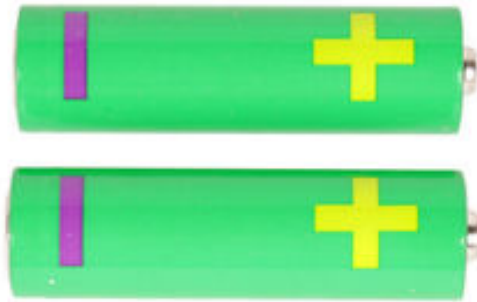
We can also connect batteries side-by-side in parallel so long as they have identical voltage ratings. When power sources are connected in parallel, the voltage remains the same, but the amount of available current increases. This is useful when a single battery does not provide enough current to power your circuit.

Keep in mind, this will only work if the batteries are the exact same voltage and should be avoided if possible. Without the proper diode protection circuit, fluctuation in voltage between the batteries will force them to try to charge one another, decreasing their lifespan.

SERIES



PARALLEL



Circuits

A circuit is a closed conductive loop in which electricity can flow freely. Basically, a circuit is a bunch of things connected together in either series or parallel that enables electricity to flow between power to ground. This can be as simple as a battery source and a motor, or as complex as a cellular phone.

In a DC circuit, power always flows between the power source (positive voltage) and ground.

In order for an electrical circuit to operate, there needs to be a path that the electricity can flow between. In fact, given the option of multiple paths, electricity will always travel the path of least resistance to ground. What this means is that electricity will always take the shortest path that offers the fewest number of obstacles. Every electronic component that creates resistance in a circuit is basically an obstacle to the flow of electricity.

Upon hearing this, you may think then that you should provide electricity with the easiest path to ground by removing these obstacles and connecting it directly. However - *and this is important to stress* - you should **NEVER** connect your positive voltage source directly to ground. Aside from the fact that removing all obstacles entirely defeats the point of electronics in the first place, this is a very bad idea.



One of the other fundamental concepts of Ohm's Law is that electricity must encounter a minimum amount of resistance in a circuit and be able to expend itself. If you connect power and ground directly together, there will be a lot of energy that has no way of expending itself. Your circuit will then try to release this unused energy in highly antisocial ways. Basically, the energy will turn into heat. However, having nothing in particular to warm, either your power source or wire will start to dramatically heat up. This can potentially result in a damaged power supply, melted wire, a fire, or potentially an explosion.

Another name of this phenomena is a "short circuit." You likely have heard this term before.

Basically, watch your power and ground connections carefully to prevent them from crossing. Don't release the circuit's "magic smoke."

